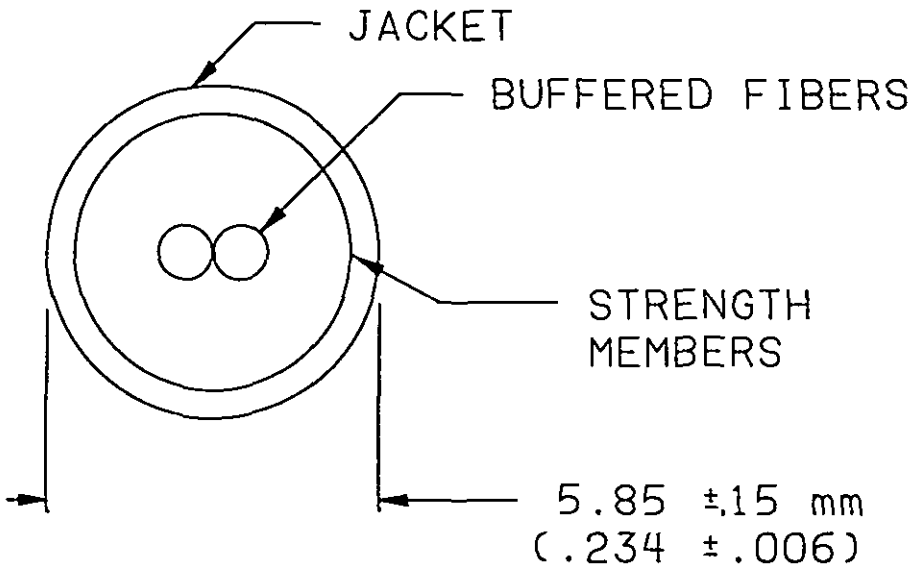
MILITARY SPECIFICATION SHEET

CABLE, FIBER OPTIC, RUGGEDIZED, RADIATION HARDENED (METRIC)

This specification is approved for use within the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of DOD-C-85045.



NOTES:
1. Dimensions are in millimeters.
2. Inches are in parentheses.

FIGURE 1. Two-fiber cable.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
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REQUIREMENTS:

Dimensions:

Cable: See figure 1.

Fiber: Per DOD-F-49291/1. Part number D49291/1-01.

Diameters:

Fiber buffer: 975 µm ±50 µm.

Concentricity:

Cladding/buffer: ≤14 percent.

Materials:

Optical fiber:

Coating: Polymer.

Buffer: Polymer elastomer, tight construction.

Strength members:

Polyarylamide, type PAA (Aramid).
Impregnated glass rods (if required).

Jackets: Flame retardant polyurethane, or equivalent.

Cable nuclear vulnerability: Dosage and recovery as specified in CR-CS-0051-001, appendix I (confidential). Test is conducted for 1,290 nm wavelength only and on primary coated optical fibers only.

Color: Black.

Cable mass per unit length: 32 kg/km maximum.

Cable continuous length:

1,025 m ±20 m.
315 m ±10 m.

Performance:

Attenuation rate:

Nonnuclear:

1.5 dB/km at +25°C and +71°C at 1,290 nm ±10 nm.
2.0 dB/km at -46°C at 1,290 nm ±10 nm.

Nuclear: 4.0 dB/km at -46°C, 3.0 dB/km at +25°C and 2.5 dB/km at +71°C at 1,290 nm ±10 nm.

Optical fiber bandwidth: ≥400 MHz-km at 1,290 nm ±10 nm.

Numerical aperture: 0.23 ±.02.

Storage temperatures: -57°C, +85°C.

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Barometric pressure (reduced):

Operating: 3,050 m.
Nonoperating: 12,200 m.

Jacket opacity: Not applicable.

Fluid immersion: Not applicable.

Flammability: 60° angle procedure.

Corner bend: Not applicable.

Knot: Force = 300 newtons.

Minimum bend radius: 30 mm.

Number of fiber splices allowed per km: None.

Life at elevated temperature: 240 hours at +110°C.

Temperature cycling shall be as specified below:

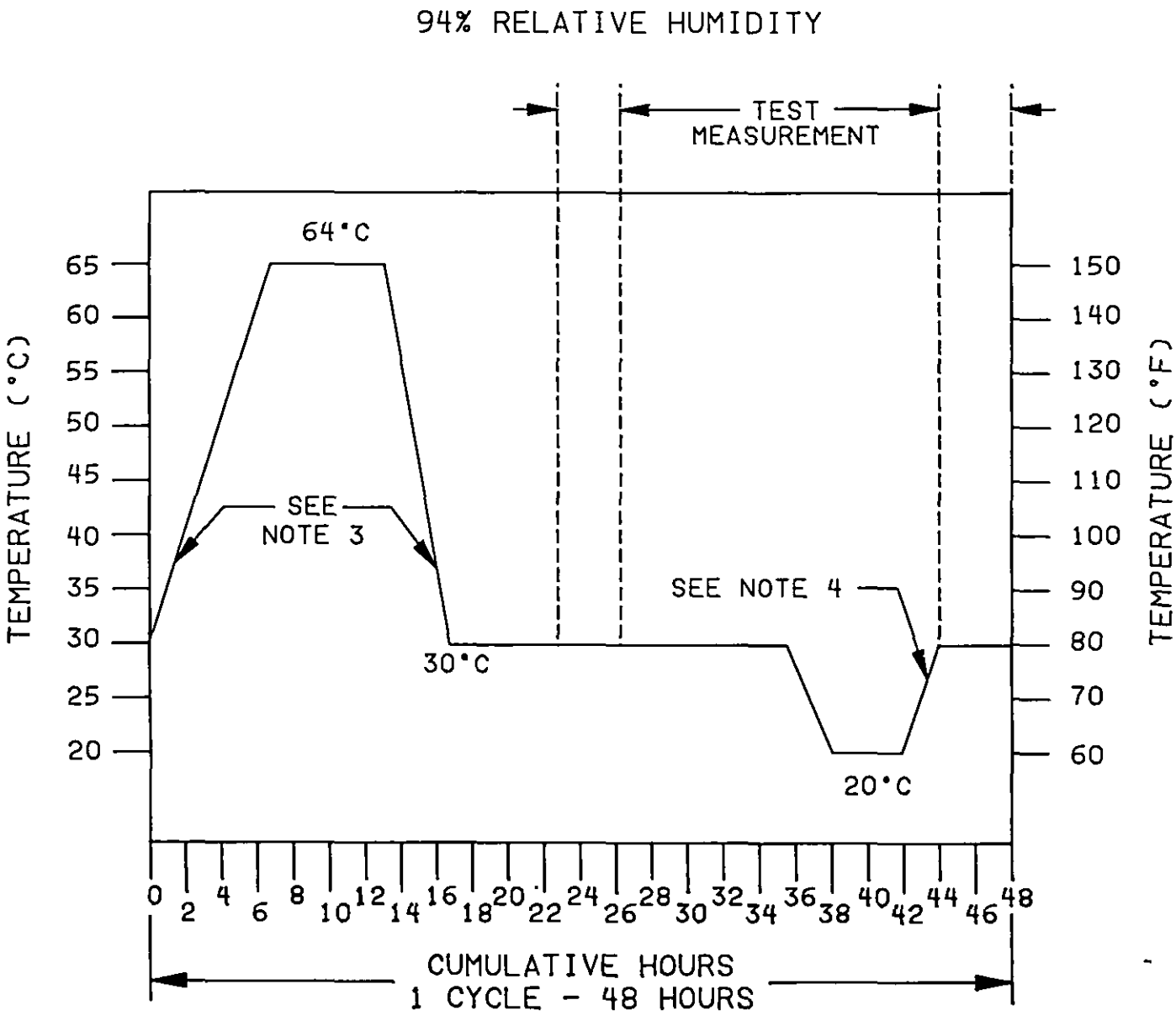
Step		Temperature (°C)	Duration (hours)
1.		Room ambient	24
2.	Reduce to	-46°C ±2°C	2
3.	Hold		8
4.	Increase to	+25°C ±2°C	2
5.	Hold		6
6.	Increase to	+71°C ±2°C	1
7.	Hold		6
8.	Reduce to	+25°C ±2°C	1
9.	Hold		6
10.	Repeat steps 2 through 9 four additional times for a total of five cycles. Induced attenuation measurements shall be taken during the last 15 minutes of steps 1, 3, 5, 7 and 9 for each cycle.		

Humidity: Exposure cycle for humidity shall be as specified on figure 2.

Impact: The impact resistance property shall be determined by the ability of the cable to meet the induced attenuation requirements.

Part number: D85045/882A.

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NOTES:

1. Tolerance during temperature change shall be not greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ±4 percent at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85 percent.
3. Rate of temperature change between 30°C and 65°C (86°F and 149°F) shall be not less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion of the curve shall be not less than 10°C (18°F).
5. Test measurements shall be taken only at the period specified in the applicable equipment or system specification.

FIGURE 2. Humidity cycling.

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Qualification inspection shall consist of performing the inspections and optical tests specified in Table I herein, in the sequence shown.

Group A inspection shall consist of performing the inspections and optical tests specified in Table II herein, in the sequence shown.

Group B inspection shall consist of performing the inspections and optical tests specified in Table III herein, in the sequence shown.

Group C inspection shall consist of performing the inspections and optical tests specified in Table IV herein, in the sequence shown.

TABLE I. Qualification Inspection 1/ 2/

Inspection	Specimen length	Sample quantity
Group I		
Visual and mechanical	1 km	6 units
Attenuation rate	1 km	6 units
Group II		
Optical fiber bandwidth	1 km	6 units
Crosstalk	1 km	6 units
Numerical aperture	2 m	3 units (1 each from 3 cables)
Group III		
Temperature cycling	1 km	6 units (3 on reel, 3 off reel)
Humidity	1 km	2 units
Storage temperature	1 km	2 units (1 on reel, 1 off reel)
Low temperature flexibility (cold bend)	8 m	3 units (1 each from 3 cables)
Cyclic flexing	5 m	6 units 3/
Crush	5 m	6 units 3/
Cable twist-bending	5 m	6 units 3/
Impact	5 m	6 units 3/
Barometric pressure (reduced)	1 km	1 unit
Life (elevated temperature)	500 m	2 units

See footnotes at end of table

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TABLE. I. Qualification Inspection - Continued.1/ 2/

Inspection	Specimen length	Sample quantity
Group III - Continued		
Tensile loading and elongation	150 m	6 units (2 each from 3 cables)
Operational tensile loading	150 m	2 units (1 each from 2 cables)
Freezing water immersion (ice crush)	30 m	2 units (1 each from 2 cables)
Fungus	2 m	2 units (1 each from 2 cables)
Knot	5 m	2 units
Cable element removability	2 m	3 units (1 each from 3 cables)
Durability of identification marking	2 m	3 units (1 each from 3 cables)
Flammability	0.6 m	3 units (1 each from 3 cables)

1/ Tolerance on 1 km lengths is plus or minus 2 percent provided results are normalized to 1 km.
2/ Tolerance on shorter lengths is plus or minus 10 percent.
3/ 3 units each from 2 cables, 2 units for each test temperature, 6 units total.

Group A sampling plan: Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be 1 percent major, 4 percent minor.

Group C sampling plan: See table IV.

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TABLE II. Group A inspection.

Inspection	Specimen length
Visual and mechanical	1 km
Attenuation rate	1 km

TABLE III. Group B inspection.

Inspection	Specimen length
Optical fiber bandwidth	1 km
Crosstalk	1 km
Numerical aperture	2 m

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TABLE IV. Group C inspection 1/ 2/.

Inspection	Inspection Interval	Specimen length	Sample quantity
Subgroup 1	(3 month intervals)		
Temperature cycling		1 km	4 units (2 on reel, 2 off reel)
Subgroup 2	(6 month intervals)		
Humidity		1 km	2 units
Storage Temperature		1 km	2 units (1 on reel, 1 off reel)
Low temperature flexibility (cold bend)		8 m	2 units
Cyclic flexing		5 m	6 units 4/
Crush		5 m	6 units 4/
Cable twist-bending		5 m	6 untis 4/
Impact		5 m	6 units 4/
Subgroup 3	(12 month intervals)		
Barometric pressure (reduced)		1 km	2 units
Life (elevated temperature)		500 m	2 units
Tensile loading and elongation		150 m	2 units
Operational temsile loading		150 m	2 units
Freezing water immersion (ice crush)		30 m	2 units
Fungus 3/		2 m	2 units
Knot		5 m	2 units
Cable element removability		2 m	2 units
Durability of identification marking		2 m	2 units
Flammability		0.6 m	2 units

1/ Tolerance on 1 km lengths is plus or minus 2 percent provided results are normalized to 1 km.
2/ Tolerance on shorter lengths is plus or minus 10 percent.
3/ Fungus resistance may be certified for quality assurance purposes.
4/ 3 units each from 2 cables, 2 units for each test temperature, 6 units total.

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Custodian:
Army - CR

Review activities:
Army - SC
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6015-A022-01)

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